

Multiple Choice (10 marks)

1. What is the greatest common divisor of $2^{1001} - 1$ and $2^{1012} - 1$?
 - (a) 1
 - (b) 2049
 - (c) 2048
 - (d) 2047
2. If x and y are directly proportional and $x = 3$ when $y = 8$, what is the value of x when $y = 13$?
 - (a) 0.615
 - (b) 4.875
 - (c) 15
 - (d) 34.667
3. Which of the following points is NOT a solution to both the inequalities $y > 9x - 8$ and $y < -x + 8$?
 - (a) $(-2, 15)$
 - (b) $(-1, 5)$
 - (c) $(0, 0)$
 - (d) $(1, 5)$
4. A hypothesis test is set up so that $P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true}) = 0.05$ and $P(\text{failing to reject } H_0 \text{ when } H_0 \text{ is false}) = 0.26$. What is the power of the test?
 - (a) 0.26
 - (b) 0.05
 - (c) 0.95
 - (d) 0.74
5. Find the value of the expression $-16 + 13 + (-33)$.
 - (a) 4
 - (b) 62
 - (c) 36
 - (d) -36

True / False (10 marks)

Answer True or False

6. True or False: The decimal representation of 7 over 33 is 0.22 repeating.
7. True or False: The value of $\frac{-\frac{18}{7}}{-\frac{6}{5}}$ is $\frac{7}{15}$ \$.
8. True or False: The numbers 1 over 8, 1 over 3, 0.75, and 5 over 6 are arranged in order from least to greatest.
9. True or False: The square of 15 is 225, which is obtained by multiplying 15 by itself.
10. True or False: The probability that the two numbers rolled on two fair 6-sided dice sum to 4 is $\frac{1}{4}$ or $\frac{1}{3}$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. George has a budget of 23 for art supplies, including markers, paper, and glue. If the combined cost of markers and paper is 15, formulate the inequality for the remaining amount he can spend on glue and explain your reasoning.
12. Abby drives 18 miles each day for her round trip to work. Calculate the total miles she drives in 5 days and discuss how rounding affects your estimation of this total.

End of Paper